

Plaksha Dynamics and Simulation

Final Exam

Thursday 8 May 2025 4 PM - 6 PM (or, up to 7 PM)

No calculators, books or notes allowed.

4 graded Problems, 120 minutes (+ 60 minutes over time)

*** How to get the highest score? ***

Please do these things:

- **Your written work.** Start each problem on a clean sheet.
— Put ***your name and *problem number and *page number** on the top of every side of every sheet.
- Draw **Free body diagrams** whenever force, moment, linear momentum, or angular momentum balance are used.
- Use correct **vector notation**.
- A+ Be (I) neat, (II) clear and (III) well organized.
- TIDILY REDUCE and box in your answers (Don't leave simplifiable algebraic expressions).
- >> Make appropriate `Matlab` code clear and correct.
You can use shortcut notation like " $\phi_7 = 2\pi$ " instead of, say, "`phi (7) = 2*pi;`".
Small syntax errors will have small penalties.
- ↗ Clearly **define** any needed dimensions ($\ell, h, d, \dots$), coordinates ($x, y, r, \theta, \dots$), variables ($v, m, t, \dots$), base vectors ($\hat{i}, \hat{j}, \hat{e}_r, \hat{e}_\theta, \hat{\lambda}, \hat{n}, \dots$) and signs ($\pm$) with sketches, equations or words.
- **Justify** your results so I can distinguish an informed answer from a guess. If you quote a fact that a grader might doubt your understanding of, explain it. Especially if it is not commonly used.
- 👁 If a problem seems *poorly defined*, clearly state any reasonable assumptions (that do not oversimplify the problem).
- ≈ Work for **partial credit** (from 60–100%, depending on the problem)
 - Put your answer is in terms of well defined variables even if you have not substituted in the numerical values.
 - Reduce the problem to a clearly defined set of equations to solve.
 - Provide computer code, or pseudo-code which would generate the desired answer, and explain the nature of the output (unless specifically precluded).

Problem 1: ____/50

Problem 2: ____/50

Problem 3: ____/50

Problem 4: ____/50

1) Simple pendulum in 2D using DAEs. A 2D rigid object swings from a hinge. There is gravity and there is no bearing friction and no air resistance. Develop pseudo code for the equations of motion using the **DAE** approach the input to the code should be the dynamic state and the output should be the rate of change of all state variables as well as the constraint force(s). With a well-labeled sketch indicate and define all parameters, points, coordinates and all dynamic state variables.

2) Central force, 2D $F(r)$ is the central force, and only force, acting on a particle with mass m in circular motion.

a) Find $F(r)$ so that the period of the motion is T (a given constant) no matter what the radius.

b) Find $F(r)$ so that the speed of the particle is v^* (a given constant) no matter what the radius.

3) Chaplygin Sleigh, 2D A rigid object slides on a horizontal plane. It contacts the ground with a skate (attached to the object, can slide freely in one direction, can rotate freely on the ground, cannot slide perpendicular to the skate). It is also kept level by sliding on frictionless pads. Define any needed coordinates, points, parameters or dynamic variables. Find the Equations of motion by any means you please.

4) particle on a wire, 2D, Lagrange equations.. A vaguely parabola -looking rigid wire has the shape $y = s^2$ where s is arc-length along the wire starting at the point $x = y = 0$. Gravity points in the $-y$ direction. A mass m slides frictionlessly on the wire. Using s as the generalized coordinate, use Lagrange equations to find the equations of motion on this wire. (Hint, or tease: this is, in disguise, one of the most famous mechanics systems).